

Introduction to Data Science

Group assignment

Measuring progress towards the UN Sustainable Development Goal 8: Decent work and economic growth

The United Nations Sustainable Development Goals (SDGs) are targets for global development that were adopted in 2015. All countries have agreed to work towards achieving them by 2030.

In this group assignment, you will assess different continents' progress towards meeting the UN Sustainable Development Goal 8: Decent Work and Economic Growth. To promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.

You are given the following data sets in csv files:

- 1. continents-according-to-our-world-in-data.csv**
 - This data set classifies countries into seven different continents: Asia, Africa, North America, South America, Antarctica, Europe, or Oceania.
- 2. gdp-per-capita-worldbank.csv**
 - This data set contains the GDP per capita, reported in 2017 USD\$, using the purchasing power parity (PPP) exchange rate for all countries in the world, going back to 1990 (if the data is available).
- 3. youth-not-in-education-employment-training.csv**
 - This data set contains the share of youth not in education, employment or training, reported as a % of the total youth population for all countries in the world, going back to 1976 (if the data is available). Youth is considered to be people of ages 15 to 24.

Assignment brief

- Perform exploratory data analysis to assess how 6 continents in the world (all continents except Antarctica) are faring with respect to the following specific targets set out by the UN:

1. *Sustain per capita economic growth in accordance with national circumstances and, in particular, at least 7 per cent gross domestic product growth per annum in the least developed countries.*
 2. *By 2020, substantially reduce the proportion of youth not in employment, education or training.*
- Locate, in collaboration with your group, an additional CSV file which can add to your analysis. This CSV file should add further information which you can incorporate into your dataset. The provenance of the CSV file should be assessed and discussed.

Assignment format and handin

- Use Github to share your code for the group project. Your report will be handed in by sharing the link to your Github repository. If you prefer, you can keep your repository private and add the instructor as a collaborator.
- Your Github repository should show evidence of collaboration by the entire team (at least one commit or pull request from each member of the team).
- Your report should state the Github usernames of your team.
- Your analysis should draw together information from the three CSV files provided as well as from the supplemental CSV file located by your team.
- Present your findings in a presentation recording of no more than 15 minutes in length and a report of no more than 5 A4 pages in length and 2500 words. Not all group members need to present in the video.
- If you use external help or resources, you must cite them and include the full bibliography in a separate References section using the Harvard referencing style. You should put any References on separate pages at the end of your report (this does not count towards the 5 page limit).
- Clean and annotate R codes used to generate the results in your presentation using R markdown and submit the Rmd file, together with the compiled pdf. This file needs to be error-free, and compilable from scratch (i.e., starting from the original .csv files). You will learn about R markdown in the coming weeks, so until then you may want to just try to keep a clean R script (e.g., using # to add comments).
- Large language models (LLMs) produce vague, inconsistent and often incorrect output. The use of LLMs is not prohibited, however work which is vague, inconsistent or incorrect will be marked accordingly.
- Your work will be marked in accordance with the Imperial undergraduate marking guidelines.
- Submit your video as an .mp4 file via the Hub, and your written report and your .Rmd code together with its compiled PDF via Github.

Guidelines

- Think what each target means carefully. There may not be one “correct” measure, so you may want to explore more than one to assess each target.
- You may wish to compare across the continents, as well as assess their individual progress across time. You may also wish to further divide up a continent (e.g., into more developed vs. less developed countries).
- Explain what data processing or analysis you did, and why.

- Try to complete all analysis early and focus on the presentation, report and cleaning the codes, as putting analysis together takes time.
- Work steadily. You would need to devote 2-4 hours each week leading up to the deadline on the assignment. Do not leave this assignment to the last minute.
- Try to use the R functions you learnt in class first, before trying to find fancier functions. As we will continue to cover more R in class, revisit the assignment with the new knowledge each week.
- Structure your report to have: 1. Introduction 2. Methodology 3. Results 4. Discussion and 5. Conclusion. You can use a similar structure for your video presentation as well.

Mark Breakdown:

	Details	Marks
Video presentation: Assessment of Target 1	Assessment of Target 1 - Analysis and results are presented in clear and meaningful ways (15) Assessment of Target 2 - Analysis and results are presented in clear and meaningful ways (15)	30
Written report: Assessment of Target 1	- Data sets are analysed in sufficient detail to assess the targets (5) - Any data processing/analysis are well-justified and thought-through (10) - Analysis is written in clear and meaningful ways (8) - Github has been used as specified to collaborate on code; visible contributions from all team members.(2)	25
Written report: Assessment of Target 2	- Data sets are analysed in sufficient detail to assess the targets (5) - Any data processing/analysis are well-justified and thought-through (10) - Analysis is written in clear and meaningful ways (10)	20
Code	- Code compiles without error (5)	

	- Code is presented and explained clearly and concisely (15)	20
References	Lack of clear citations of any resources used will result in marks being deducted. In extreme cases, the assignment may be flagged for plagiarism.	5
Total		100